Graph Neural Networks: An Orientation

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Goal

- This course is designed to provide an in-depth exploration of Graph Neural Networks (GNNs), one of the most exciting areas of recent developments in machine learning and AI research. Students will gain a comprehensive understanding of GNNs, including their foundations, state-of-the-art models, and practical applications.

- We will start with the basics of graph theory and Graph Convolutional Networks (GCNs), then move into more advanced topics, including Graph Attention Networks (GATs), structure-preserving GNNs, overcoming GCN limitations with models like GCNII, and Graph Transformer Networks. The course will also cover graph pooling techniques, graph autoencoders, and generative models for graphs. Moreover, we will explore handling dynamic graphs and heterogeneous graphs.

- Throughout the course, we will place emphasis on the practical application of these models, focusing on various real-world scenarios and use cases. The course will mix theoretical lectures with practical exercises and coding assignments, giving students a balance between understanding the principles behind GNNs and applying them in practice.

Content

- By the end of the course, students will be able to:
  - Understand the foundational concepts of graph theory and how they apply to machine learning.
  - Understand the architecture, training, and use of various Graph Neural Network models, including GCNs, GATs, and Graph Transformers.
  - Develop and train GNNs to solve real-world problems.
  - Recognize the limitations of current GNNs and understand the solutions offered by state-of-the-art models.
  - Analyze, evaluate, and compare different GNN models based on their performance, applicability, and limitations.
  - Stay up-to-date with the latest research and developments in the field of GNNs.
  - Apply graph neural networks to different fields, such as natural language processing, social network analysis, bioinformatics, and recommendation systems.
- By the end of this course, students should be well-equipped to implement GNNs in their projects or research, as well as understand the current research landscape in the field of Graph Neural Networks.

2. Reference Book

- Students do not need to buy textbooks.
- Most of the lecture materials will be provided in digital format (e.g., PDF files).

Textbooks

- Lingfei Wu, Peng Cui, Jian Pei, Liang Zhao, “Graph Neural Networks: Foundations, Frontiers, and Applications”, Springer, 2022.
- Zhiyuan Liu, Jie Zhou, “Introduction to Graph Neural Networks”, Springer, 2020.

3. Course Schedule

- **Week 1: Introduction to Graphs**
  - Overview about graph theory, types of graphs and their applications.
  - Basic graph concepts (nodes, edges, degree, etc.) and graph representation (adjacency and sparse matrix).
  - Centrality measurement: Degree centrality, Betweenness centrality, Closeness centrality, Eigenvector centrality, and PageRank.
  - Graph kernel.

- **Week 2: Node Embedding and Shallow Models in Graphs**
  - Overview node embedding, edge embedding, and graph embedding.
  - Shallow models: Random, proximity, and heterogeneous – based models.

- **Week 3: Message Passing Mechanism & Graph Neural Network**
  - GNN overview and motivation.
  - Basic framework of Graph Convolution Networks and convolution operation in GCNs.

3. Course Schedule

- **Week 4: Scalability of Graph Neural Networks**
  - Issues towards the large-scale GNNs.
  - Node-wise sampling with GraphSage.
  - Graph-wise sampling with ClusterGCN
  - GraphSAIN
- **Week 5: Training Deeper GNNs**
  - Why do we need deeper GNNs?
  - What impedes GNNs to go deeper?
  - Solutions for alleviating over-smoothing and over-fitting
- **Week 6: Attentive Graph Neural Networks and Graph Pooling**
  - Graph neural network issues.
  - Attention in Graph neural networks.
  - Attention in Heterogeneous graphs.
  - Graph Pooling

3. Course Schedule

- **Week 7: Structure-Preserving Graph Neural Networks**
  - Computational graph.
  - Graph Isomorphism and GNN problems.
  - Expressive power of GNNs.
  - How to build the most powerful GNNs?

- **Week 8: Midterm Exam**

- **Week 9: High-order Graph Neural Networks**
  - Message Passing Neural Networks (MPNNs) issues.
  - MPNNs and Color refinement.
  - High-order MPNNs.

3. Course Schedule

- **Week 10: Graph Transformers**
  - From Message Passing to Self-attention.
  - Positional Encoding in Graphs.
  - Transformers are Graph Neural Networks.
  - Representative Transformer models

- **Week 11: Spatio-temporal Graph Neural Networks (STGNN)**
  - Introduction STGNN.
  - Application in Action Recognition.

- **Week 12: Graph Neural Network for Recommendation Systems**
  - Tasks and Evaluation Metrics for GNNs in Recommendation Systems.
  - Embedding models are effective for Recommendation Systems.
  - GNNs for Recommendation Systems.

3. Course Schedule

- **Week 13: Graph Neural Networks for Molecular Structure Learning**
  - Introduction to GNNs and Molecules.
  - Molecules represented as Graphs.
  - GNNs for molecular learning tasks.
  - Equivariant GNNs
  - Recent Advances in GNNs on Molecular Learning

- **Week 14: Heterogeneous Graph Neural Networks and Knowledge Graph**
  - From Homogeneous Graphs to Heterogeneous Graphs.
  - Knowledge Graphs.
  - Knowledge Graph Representation Learning

- **Week 15: Consolation**
- **Week 16: Final Exam**

4. Assessment Criteria and Grading Scale

Assessment

- Midterm exam: 30%.
- Final exam: 30%.
- Homework: 30%.
- Attendance: 10%.
- All test questions are made up within lecture slides and assignments.

Tentative Grading Scale

- A: 80 - 100.
- B: 60 - 80.
- C: ~60.
- F: Absence from exams.

Weekly Programming Practice

- Perform practical programming in one class every week.
- Do assignment after practice class every week.

6. Instructor Information

O-Joun Lee (이오준)

- Email: ojlee@catholic.ac.kr.
- Office: T404.
- Office phone number: 5516.

- Office hour: visit office anytime.
- Please feel free to contact me for any class issues

7. Teaching Assistant Information

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